Current status of the DY absolute cross-section measurements study

Chatura Kuruppu

New Mexico State University

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Outline

1. Events from Runs 2 and 3 (Input Files Used)
2. Event Selection Used (DocDB 2111-V42 & Fiducial Cuts)
3. New Reconstruction Efficiency Correction (D1-10)
4. Latest Vs Existing Reconstruction Efficiencies (LH2)
5. Latest Vs Existing Reconstruction Efficiencies (LD2)
6. Acceptance Corrections (LH2 & LD2)
7. Cross-Section Comparison All Roadsets (LH2 & LD2)

Events from runs 2 and 3 (Input Files Used)

Note: Currently using all the runs saved in runs 2 and 3.

Harsha's ROOT files:
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/:

- merged_RS57_LH2_1_1138.root
- merged_RS57_LD2_3_1138.root
- merged_RS57_Empty_2_1138.root
- merged_RS59_LH2_1_465.root
- merged_RS59_LD2_3_466.root
- merged_RS59_Empty_2_466.root
- merged_RS62_LH2_1_1234.root
- merged_RS62_LD2_3_1237.root
- merged_RS62_Empty_2_1234.root
- merged_RS70_LH2_1_264.root
- merged_RS70_LD2_3_266.root
- merged_RS70_Empty_2_267.root

Abi's ROOT files:
/seaquest/users/apun/e906_projects/rs67_merged_files/:

- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root

Event Selection Used: Chuck Cuts (DocDB 2111-V42) & Fiducial Cuts

Chuck Cuts (DocDB2111-V42) with beam_offset $= 1.6$ cm and mass cut $= 4.2$ GeV/c^2:

Dimuon Cuts (dimuon_cut_2111_v42)

- **Vertex:** $|dx| < 0.25$, $|dy - 1.6| < 0.22$, $-280 < dz < -5$ cm
- **Transverse Vertex:** $dx^2 + (dy - 1.6)^2 < 0.06$ cm^2
- **Momentum:** $|dp_x| < 1.8$, $|dp_y| < 2.0$, $dp_x^2 + dp_y^2 < 5.0$ (GeV/c)^2, $38 < dp_z < 116$ GeV/c
- **Kinematics:** $4.2 <$ mass < 8.8 GeV/c^2, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, $|\cos\theta| < 0.5$
- **Fit & Separation:** |trackSeparation| < 270 cm, $\chi^2_{\text{dimuon}} < 18$

Single Track Cuts (track1_cut & track2_cut, $i = 1, 2$)

- **Target/Dump** χ^2: $\chi^2_{i,\text{target}} < 15$, $\chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{upstream}}$, $\chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{dump}}$
- **Track Quality:** nHits$_i > 13$, $\chi^2_i/(\text{nHits}_i - 5) < 12$, $-320 < z_{i,v} < -5$ cm
- **Apertures:** $x_{i,t}^2 + (y_{i,t} - 1.6)^2 < 320$, $16 < x_{i,d}^2 + (y_{i,d} - 1.6)^2 < 1100$ cm^2
- **Station 1 & 3:** $9 < p_{z,i}^{\text{st1}} < 75$, $|p_{y,i}^{\text{st1}}| > 0.02$ GeV/c, $y_{i,\text{st1}}/y_{i,\text{st3}} < 1$, $y_{i,\text{st1}} \cdot y_{i,\text{st3}} > 0$
- **Δp Matching:** $||p_{x,i}^{\text{st1}} - p_{x,i}^{\text{st3}}| - 0.416| < 0.008$, $|p_{y,i}^{\text{st1}} - p_{y,i}^{\text{st3}}| < 0.008$, $|p_{z,i}^{\text{st1}} - p_{z,i}^{\text{st3}}| < 0.08$

Track Pair Cuts (tracks_cut_2111_v42)

- $|\chi^2_{1,\text{target}} + \chi^2_{2,\text{target}} - \chi^2_{\text{dimuon}}| < 2.0$
- $y_{1,\text{st3}} \cdot y_{2,\text{st3}} < 0$ (opposite top/bottom halves, no beam offset)
- nHits$_1$ + nHits$_2 > 29$ and nHits1St1 + nHits2St1 > 8
- $|x_{1,\text{st1}} + x_{2,\text{st1}}| < 42$ cm

Occupancy Cuts (occ_cut_2111_v42)

- $D_1 < 400$, $D_2 < 400$, $D_3 < 400$, and $D_1 + D_2 + D_3 < 1000$
- *Note:* Intensity cuts (intensity_cut_2111_v42) are not applied.

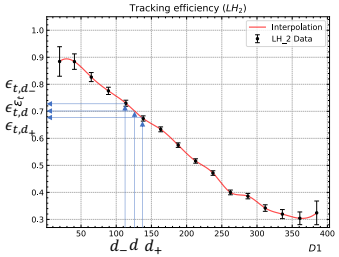
Additional Fiducial Cut Applied

In addition to the Chuck cuts above, we apply the kinematic fiducial cut:

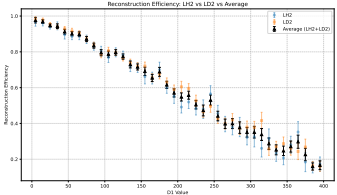
$(4.2 <$ mass < 8.8 GeV/c^2) & ($0.0 < x_F < 0.8$) & ($0 < p_T < 1.8$ GeV/c)

New Reconstruction Efficiency Correction (D1-10)

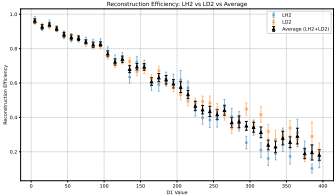
Previous Global Curve (All RS)



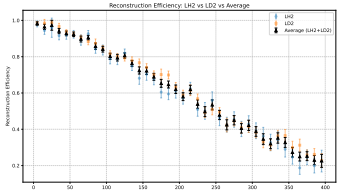
RS57 Efficiency Curve



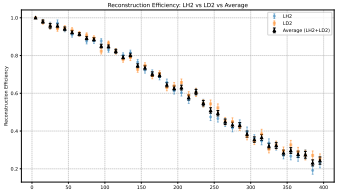
RS59 Efficiency Curve



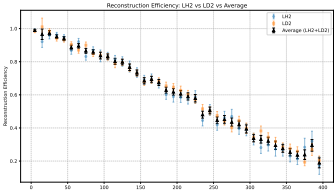
RS62 Efficiency Curve



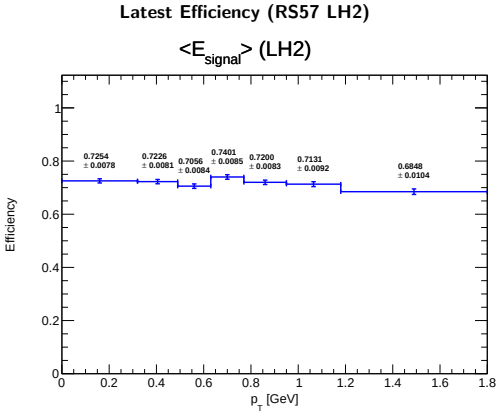
RS67 Efficiency Curve

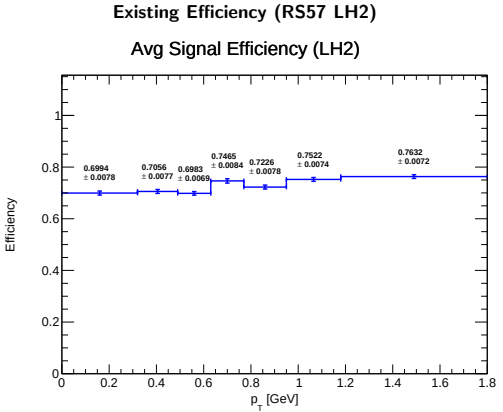


RS70 Efficiency Curve

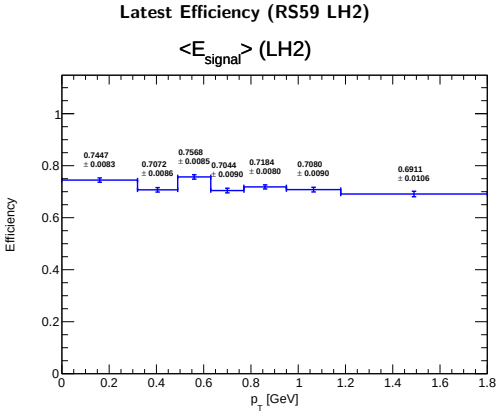


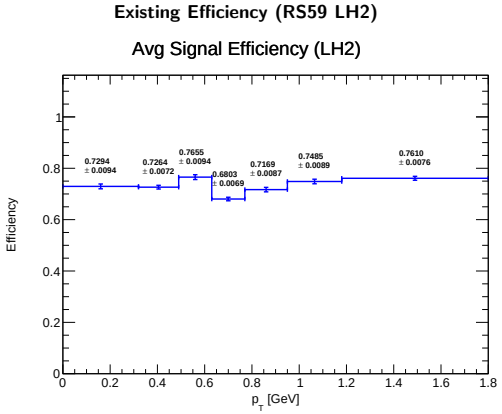
RS57: Latest Vs Existing Reconstruction Efficiencies (LH2)





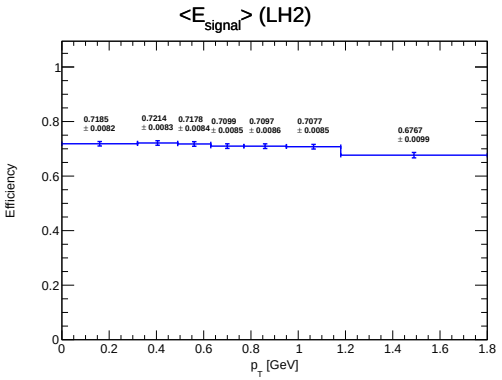
RS59: Latest Vs Existing Reconstruction Efficiencies (LH2)



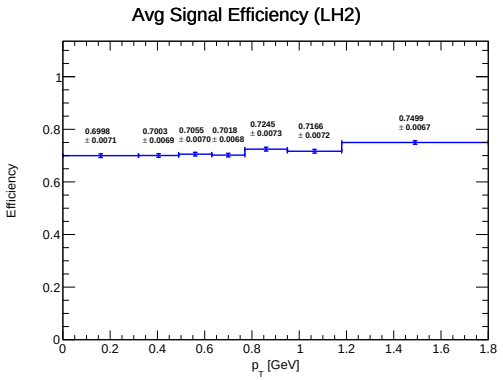


RS62: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS62 LH2)

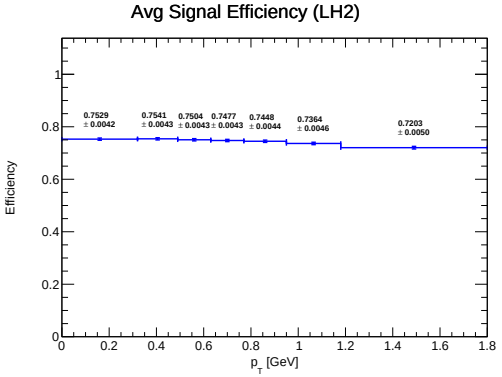


Existing Efficiency (RS62 LH2)

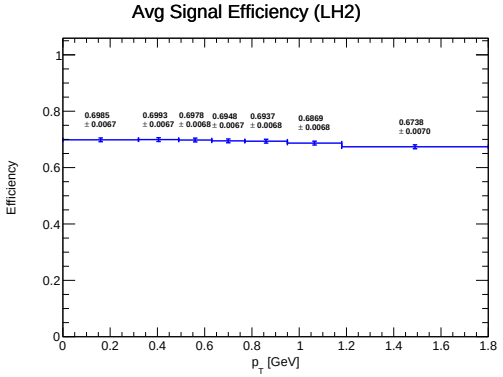


RS67: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS67 LH2)

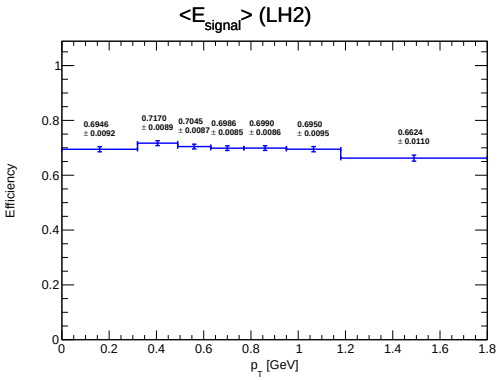


Existing Efficiency (RS67 LH2)

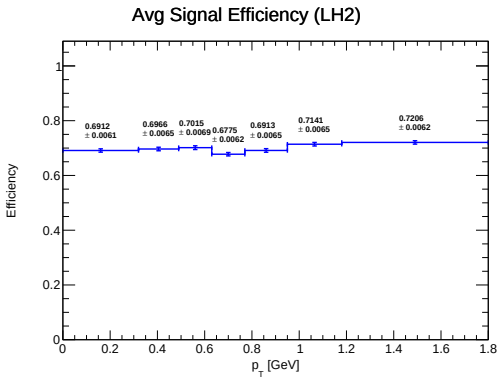


RS70: Latest Vs Existing Reconstruction Efficiencies (LH2)

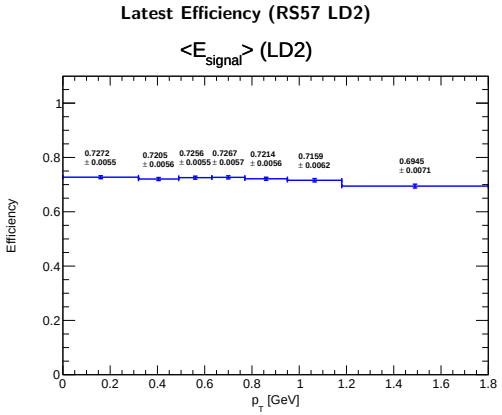
Latest Efficiency (RS70 LH2)

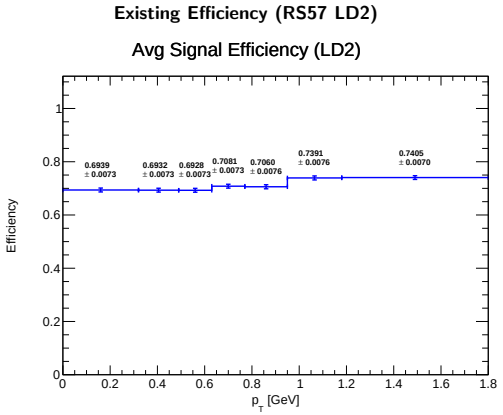


Existing Efficiency (RS70 LH2)

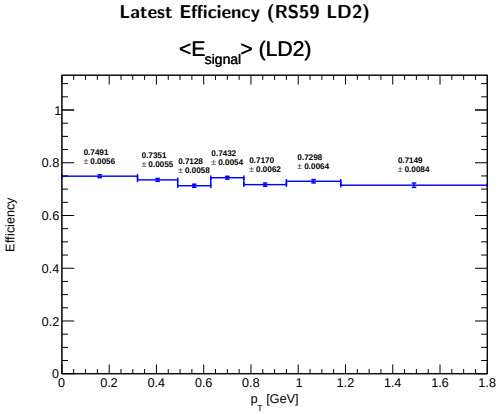


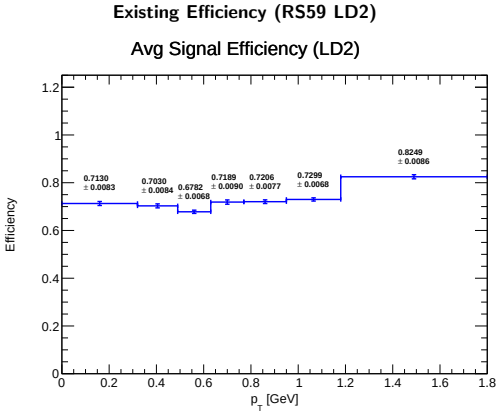
RS57: Latest Vs Existing Reconstruction Efficiencies (LD2)





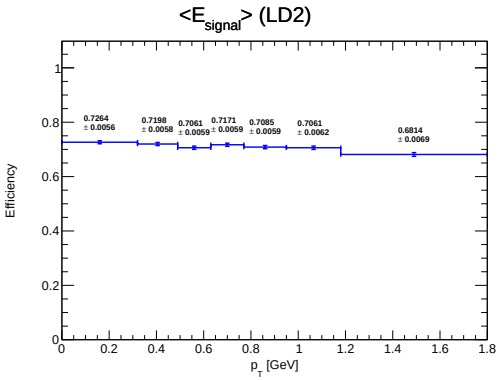
RS59: Latest Vs Existing Reconstruction Efficiencies (LD2)



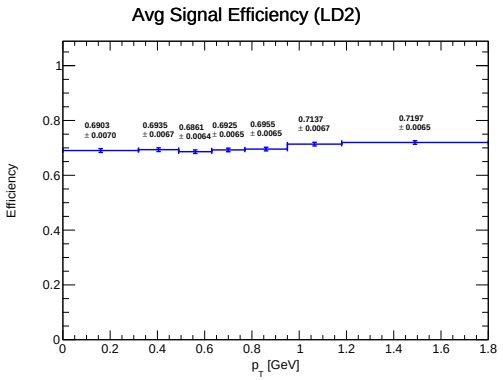


RS62: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS62 LD2)

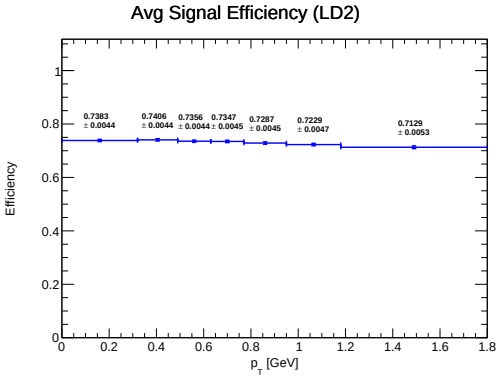


Existing Efficiency (RS62 LD2)

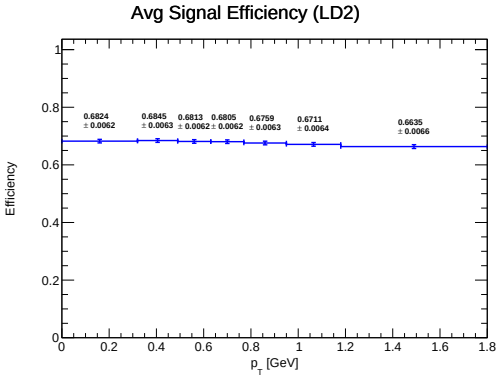


RS67: Latest Vs Existing Reconstruction Efficiencies (LD2)

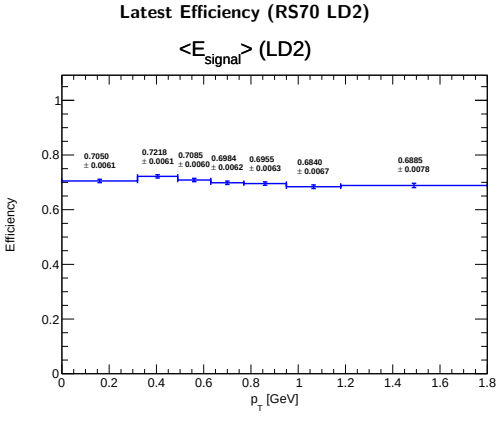
Latest Efficiency (RS67 LD2)

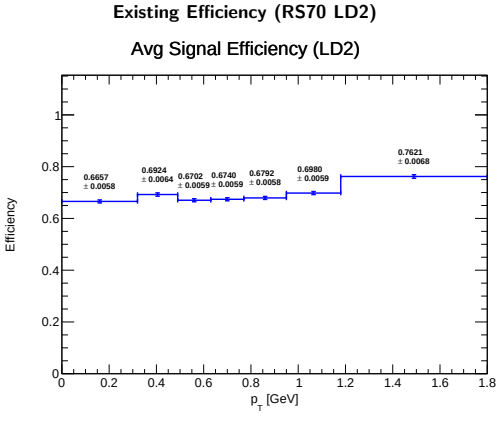


Existing Efficiency (RS67 LD2)



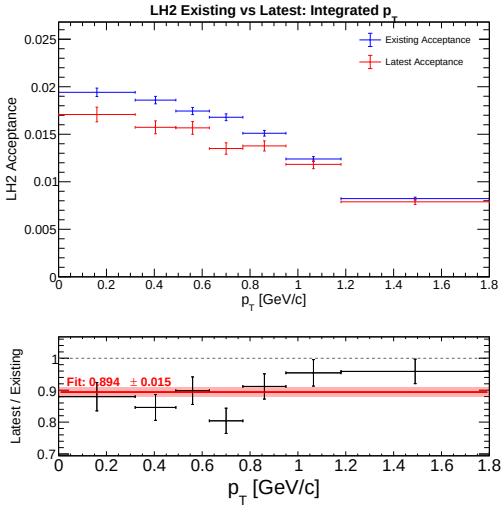
RS70: Latest Vs Existing Reconstruction Efficiencies (LD2)



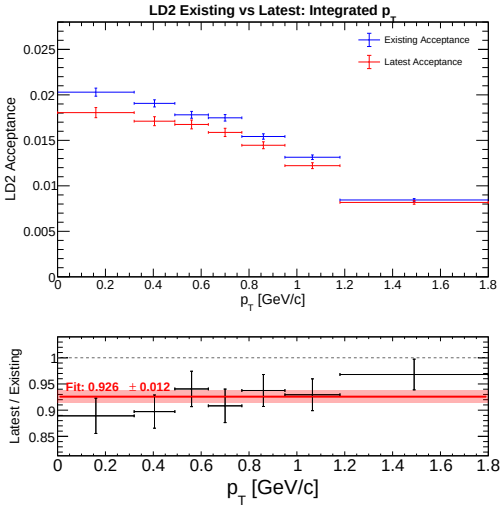


RS57 Acceptance Corrections

LH2 (RS57)

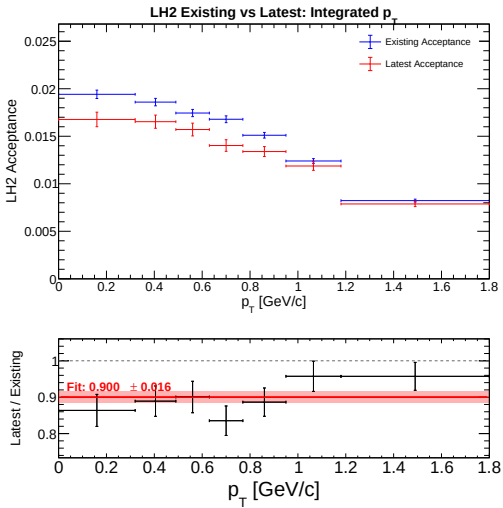


LD2 (RS57)

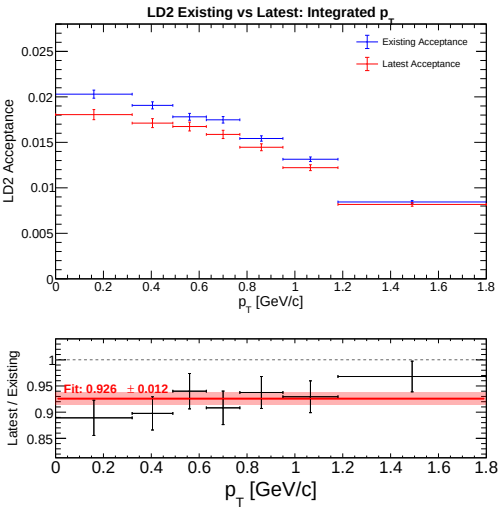


RS59 Acceptance Corrections

LH2 (RS59)

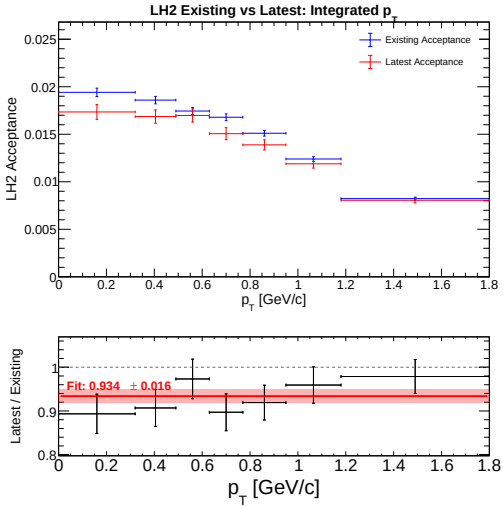


LD2 (RS59)

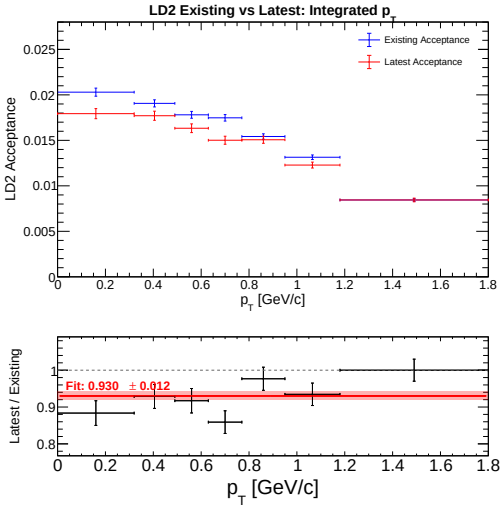


RS62 Acceptance Corrections

LH2 (RS62)

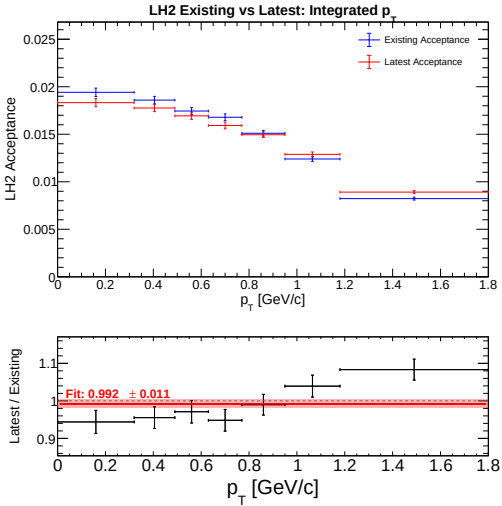


LD2 (RS62)

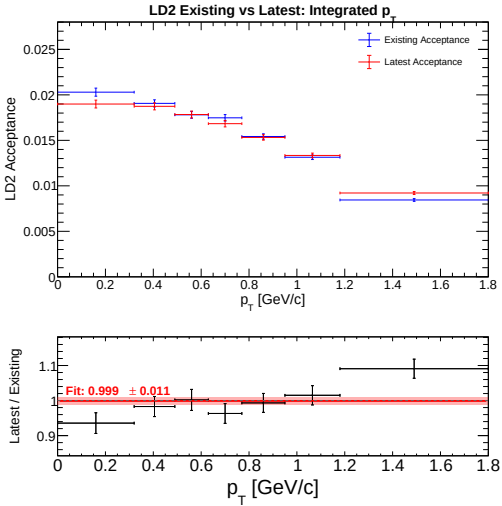


RS67 Acceptance Corrections

LH2 (RS67)

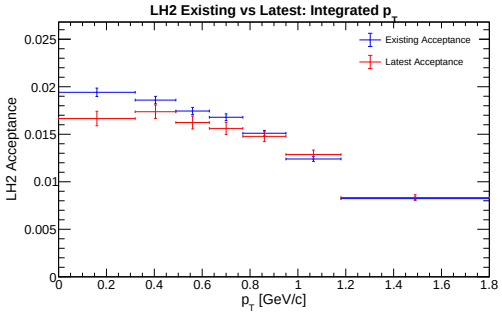


LD2 (RS67)

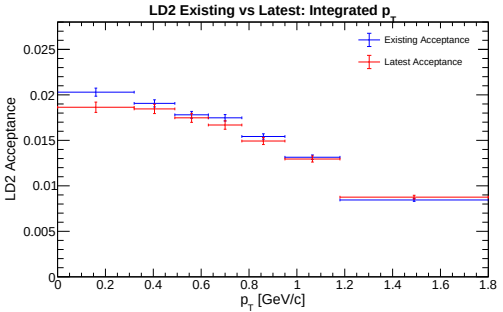


RS70 Acceptance Corrections

LH2 (RS70)

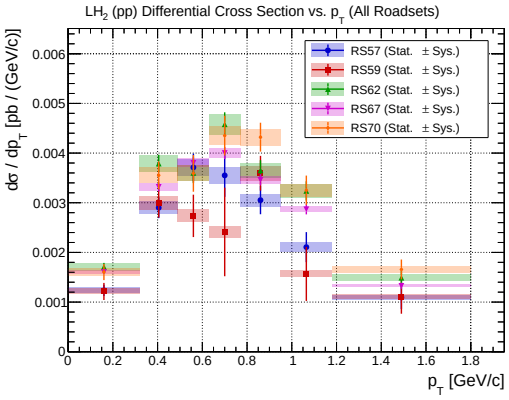


LD2 (RS70)

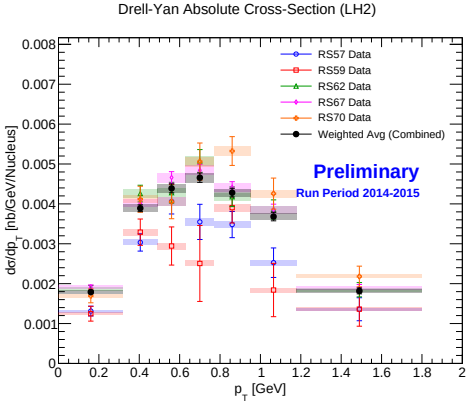


LH2 Cross-Section Comparison All Roadsets

Latest (LH2 All Roadsets)

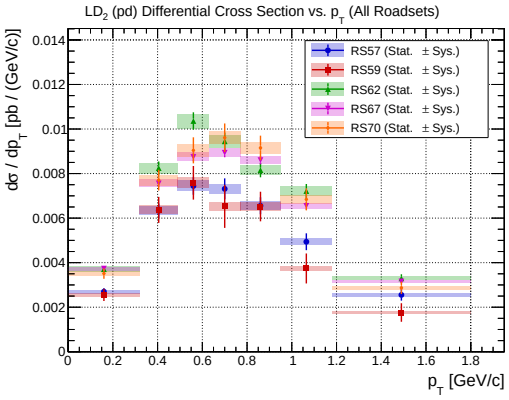


Existing (LH2 All Roadsets)



LD2 Cross-Section Comparison All Roadsets

Latest (LD2 All Roadsets)



Existing (LD2 All Roadsets)

